

# Compiling the Market Information Set: Band Content, Point-in-Time Measurement, and Economic Value in Cryptocurrency Markets

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Canonical TeX with displayed equations: pdf/sci/main\_jf\_rfs.tex. Chinese full manuscript: pdf/cn/main\_cn\_jf.md. Theory source: pdf/original/. EvoQuant is the laboratory, not the theory source.

## Abstract

Conditional asset pricing treats the investor's information set  $I_t$  as primitive. In cryptocurrency markets, asynchronous and intermittently missing evidence makes *compilation of  $I_t$*  first-order. We develop a theory of information-set compilation whose core objects are epistemic observations, a compilation operator between raw and usable filtrations, reconstruction bounds, and an SDF interface under which pricing statements are constrained to the compiled  $\sigma$ -field; leave-one-band-out (LOBO) value then decomposes into content and gating channels.

Empirically, on a PIT panel (399 days; durable bands: exchange, macro, alternative), vintage-safe macro and stablecoin-flow *content* enters a transparent action rule under a previous-close clock. The pre-specified contrast—mechanism versus same-input momentum—is positive OOS ( $\Delta CE$  0.334,  $p = 0.034$ ) and remains significant after 10–25 bps costs as a *relative* value gap, even though absolute CE is cost-fragile. LOBO losses operate mainly through content; VIX/DXY conjunction tests, joint deletion, and a scrambled-content placebo discipline the design. A 2017–2026 audit finds no unconditional edge over momentum when band archives are absent. World-quality indices and selective prediction remain part of the theory but are secondary in this sparse archive.

**Keywords:** information set; selective prediction; cryptocurrency; point-in-time; measurement error; world models. **JEL:** G12, G14, C58, C55

## 1. Introduction

Modern asset pricing conditions on an information set  $I_t$ . The literature's discipline has focused on how  $I_t$  is used—for discount factors, return prediction, and machine-learning pricing kernels—not on how  $I_t$  is compiled from asynchronous, quality-heterogeneous evidence. In cryptocurrency markets, compilation is first-order: venue fragmentation, perpetual leverage, options walls, unlocks, on-chain flows, and macro liquidity can map the same price path into incompatible states.

This paper makes three contributions, ordered by what the present archive can identify. **Theory (core apparatus):** epistemic observations, compilation operator  $\Pi_t$ , reconstruction bounds, an SDF interface on  $\blacksquare_t^{AI}$ , and LOBO content/gating decomposition; WMI/ACWMI and abstention complete the apparatus but are secondary empirics here. **Instrument:** a reproducible multi-band laboratory. **Identification (headline):** vintaged macro/stablecoin content in a transparent rule versus same-input momentum, disciplined by LOBO, conjunction tests, scrambled placebo, costs, and a 2017–2026 audit without band archives. We claim relative information value of compiled band content—not unconditional alpha and not a completed selective-prediction horse-race.

## 2. Related literature

**Information sets in asset pricing.** Classic and modern frameworks treat  $I_t$  as given (Cochrane 2005; Fama–French 1993; Gu–Kelly–Xiu 2020; Kelly–Pruitt–Su 2019; Nagel 2021; Harvey–Liu–Zhu 2016). Machine-learning asset pricing expands the feature span inside  $I_t$  but typically assumes synchronized panels. We study the prior step: compilation.

**Cryptocurrency market structure.** Fragmentation and distinctive risk factors (Makarov–Schoar 2020; Liu–Tsyvinski–Wu 2022) motivate multi-band worlds rather than price-only predictors.

**Selective prediction and measurement timing.** When the world is thin or dishonest, abstention can be optimal. Macro vintages and crypto freshness/missingness make measurement timing first-class.

### 3. Theory

This section restores the formal World-Model-First / RCA-WM apparatus. Empirics instantiate these objects; they do not replace them.

#### 3.1 World-model quality: breadth, stability, honesty

Let  $K$  be the number of critical evidence bands and  $a_{k,t} \in \{0,1\}$  an AI-usable indicator. Breadth, stability, and honesty are:

$$B_t = (1/K) \sum_k a_{k,t}, \quad U_t = \exp(-\sum_j \omega_j d_{j,t}), \quad H_t = 1 - (1/J) \sum_j m_{j,t}.$$

The production baseline World Model Index is the product form

$$WMI_t = B_t \times U_t \times H_t.$$

Hierarchical breadth and continuous honesty used in the PIT panel:

$$B_t^{\text{hier}} = 0.25 B_t^{\text{dom}} + 0.35 B_t^{\text{band}} + 0.40 B_t^{\text{asset}}, \\ H_t^{\text{cont}} = \exp(-2c_t) \max(0, 1 - 0.5(1 - e_t)),$$

where  $e_t$  is the ready-share and  $c_t$  a contamination share.

#### 3.2 Epistemic observations

An AI-consumable market observation is not a bare scalar:

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}),$$

with value, latest-available time, quality, main-view gate, and semantic role. Numerically identical  $x$  with different  $(\tau, q, g, r)$  are different world-model objects.

#### 3.3 Asynchronous state space and lag error

Latent states evolve as  $S_{t+1} = F(S_t, \eta_{t+1})$ . Source  $j$  observes a lagged map  $X_{j,t}^{\text{obs}} = h_j(S_{t-\blacksquare}) + v_{j,t}$ . If  $h_j$  is Lipschitz, reconstruction error admits a bound with delay, noise, and missingness terms:

$$\blacksquare S_t - S_t \blacksquare \leq C_1 \sum \omega_j \blacksquare_{j,t} + C_2 \sum \omega_j \blacksquare v_{j,t} + C_3 \sum \omega_j (1 - z_{j,t}).$$

Latest-snapshot, freshness/TTL, and readiness gates target these three terms.

#### 3.4 Filters and the compilation operator

$$F_t^{\text{raw}} = \sigma(\{X_{j,t}^{\text{obs}}\}), \quad F_t^{\text{AI}} = \sigma(W_t^{\text{AI}}, D_t), \quad \Pi_t = B_t \blacksquare M_t \blacksquare A_t, \quad W_t^{\text{AI}} = \Pi_t(F_t^{\text{raw}}).$$

**Proposition 1 (compilation  $\neq$  feature expansion).** Enlarging  $F^{\text{raw}}$  without a well-defined  $\Pi$  need not enlarge decision-relevant  $F^{\text{AI}}$ : ungated, stale, or role-incoherent evidence can expand raw span while shrinking usable world quality via  $H_t$  and  $U_t$ .

*Proof sketch.* Add a source  $j^*$  with large lag or missingness flag. If  $M_t$  fails to exclude it,  $H$  falls by  $\Delta > 0$  and freshness penalties weakly lower  $U$ . Breadth rises by at most  $1/K$ , but multiplicative  $WMI = BUH$  can fall when  $UH$  losses dominate. Since  $W^{\text{AI}} = \Pi(F^{\text{raw}})$  inherits gated objects, raw  $\sigma$ -field expansion need not enlarge payoff-relevant AI-visible information. ■

**Proposition 2 (SDF interface: compilation constrains pricing).** An agent whose usable conditioning information is  $F^{AI}$  can only implement pricing statements  $E[mR | F^{AI}] = 1$ . Evaluating the same agent against the finer  $F^{raw}$  produces a *compilation wedge*  $w_t = E[mR | F^{raw}] - E[mR | F^{AI}]$ , which is zero for all payoffs iff compilation loses no pricing-relevant information. Alpha measured against the raw filtration confounds skill with compilation loss.

*Proof sketch.* By iterated expectations  $E[w_t | F^{AI}] = 0$ , but  $w_t \neq 0$  pointwise unless the raw-conditional moment is  $F^{AI}$ -measurable. If a band with  $MIG > 0$  is discarded, a payoff exists whose raw-conditional expectation varies with that band while the compiled one does not. ■

**Proposition 3 (lag reconstruction bound).** Under Lipschitz observation maps and bounded increments  $|S_u - S_{u-1}| \leq \delta$ , reconstruction error admits the delay / noise / missingness bound above with  $C' = C\delta$  independent of the AI model class.

*Proof sketch.* Lipschitz gives  $|h_j(S_t) - h_j(S_{t-\tau})| \leq L_j \delta$ . Decompose  $S - S$  into lagged-map bias,  $v$ , and missingness; triangle inequality + weights  $\omega_j$  yield the bound. TTL / readiness operators shrink exactly those three terms. ■

### 3.5 ECP, MIG, and causal DAG

$$ECP_t = 1\{\text{conf}_t > c\} 1\{WMI_t < w\}.$$

Thick versus thin information sets differ by conditional mutual information; band  $k$ 's marginal information gain on task  $m$  is  $MIG_{k,t}^{(m)} = I(R_t^{(m)}; E_{k,t} | I_t^{(-k)})$ .

Identification DAG:  $O_t \rightarrow W_t \rightarrow A_t$ , with market complexity  $M_t$  and latent configuration  $C_t$  as confounders. Availability shocks  $O_t$  are the preferred quasi-exogenous lever.

### 3.6 Bayesian abstention

With action set  $A = \{\text{bullish}, \text{bearish}, \text{neutral}, \text{abstain}\}$ ,

$$a_t^* = \arg \min_a E[\ell(a, R_t) | W_t].$$

Abstain when every non-abstain action has expected loss above a world-dependent cost  $c_{abs}(W_t)$ .

**Proposition 4 (world-conditional abstention).** If  $\ell(\text{abstain}) = c_{abs}(W)$  and every non-abstain action has  $E[\ell | W] > c_{abs}(W)$ , then  $a^* = \text{abstain}$ . Under a single-crossing assumption (now formalized as Assumption 1), with non-abstain loss and  $c_{abs}$  weakly decreasing in  $WMI$ , the abstention region is a lower set in  $WMI$  (implemented by IS-frozen ACWMI thresholds).

*Proof sketch.* First claim is the Bayes argmin. Second: let  $L(W) = \min_{a \neq \text{abs}} E[\ell | W]$ ; single crossing makes  $\{W: L(W) > c_{abs}\}$  a lower contour set. ■

### 3.7 ACWMI and explanation metrics

$$ACWMI_t = \exp(\sum_i \gamma_i(r_t) \log x_{i,t} / \sum_i \gamma_i), \text{ with } x = (B^{\text{hier}}, U, H^{\text{cont}}, S, C).$$

$EAR_t = (\# \text{ evidence-bound claims}) / (\# \text{ claims})$ ,  $UCR_t = 1 - EAR_t$ ,  $EV_t = d(\Phi_t, \Phi_{t-1}) / (1 + d(W_t, W_{t-1}))$ . Thin and thick worlds also differ in explanation sets  $\Phi_t$ .

**Proposition 5 (ACWMI factor monotonicity).** With  $\gamma \geq 0$  and  $x \in (0, 1]$ ,  $\partial ACWMI / \partial x_k > 0$  and  $\log ACWMI$  is concave in  $\log x$ . A proportional honesty deterioration cannot be offset one-for-one by raising breadth with equal weight.

*Proof sketch.*  $\log ACWMI = \sum w_i \log x_i$  with  $w_i = \gamma_i / \sum \gamma$ ;  $\partial ACWMI / \partial x_k = ACWMI \cdot w_k / x_k > 0$ . Under contamination,  $\Delta \log H$  is large while  $\Delta \log B = O(1/K)$ , so net  $\sum w \Delta \log x < 0$ . ■

**Proposition 6 (LOBO as economic MIG, with channel decomposition).** Let  $V(I)$  be OOS CE of a fixed rule on information set  $I$  and  $MIG_k = V(I) - V(I \setminus E_k)$ . Under Blackwell monotonicity and non-redundancy,  $MIG_k > 0$ . Band  $k$  enters through a *content* channel (its signal changes the action) and a *gating* channel (its availability changes B/U/H and abstention); the MIG telescopes exactly into the two channel effects. In the PIT laboratory, forcing the band tilt to zero implements content deletion; setting band status to missing implements gating deletion; both together implement  $I \setminus E_k$ .

*Proof sketch.* Blackwell  $\Rightarrow V(I) \geq V(I \setminus E_k)$ ; strict when  $E_k$  has unique predictive content. The decomposition is a telescoping identity over the three experiment arms (content-only, gating-only, both). ■

**Proposition chain (summary).** Prop.1 separates raw span from usable world quality; Prop.2 constrains any pricing kernel to the compiled filtration (alpha vs compilation wedge); Prop.3 grounds freshness/TTL as reconstruction controls; Prop.4 justifies world-conditional abstention; Prop.5 makes ACWMI a strict quality index; Prop.6 maps LOBO CE drops to economic MIG and separates content from gating. Empirics below instantiate this chain—they do not replace it.

#### **4. The EvoQuant laboratory as measurement instrument**

EvoQuant supplies multi-band collectors, readiness scoring, BandPITService / time\_slice PIT reconstruction, availability-shock queries over collection\_runs, and configurable WMI/ACWMI thresholds (WORLD\_MODEL\_INDEX\_MODE, WMI\_ABSTAIN\_THRESHOLD, ACWMI\_ABSTAIN\_THRESHOLD). Epistemological order: theory first, laboratory second.

## 5. Data and real multi-band PIT archive

### 5.1 Populated archive

Exchange (OKX): ~9373 klines/merged bars, daily 2025-06-30 → 2026-08-03; ~2448 funding rows. Macro: 81013 vintaged points. Alternative: 88761 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. Analytics snapshots remain sparse historically; multi-band PIT uses raw history tables via `build_pit_archive.py` (production path prefers `BandPITService`).

### 5.2 PIT panel

Timing protocol: decision at previous close. For payoff day  $t$ , band statuses are evaluated at  $(t-1)$  23:59; content tilts use vintaged `available_at` with a one-day lag. Panel: 2025-07-01 → 2026-08-03, 399 days  $\times$  10 assets = 3990 rows. Durable bands: exchange, macro, alternative. Yahoo daily returns provide payoffs; engines use only pre- $t$  returns.

Migration disclosure: Migrated from archived same-day-asof panel because local SQLite history was empty; statuses for day  $t$  are taken from archived day  $t-1$  EOD.. When SQLite history is populated, statuses are rebuilt from raw tables; empty-history environments use the checked-in archive shifted to the same previous-close clock and never overwrite band-content caches with empty pulls.

Empirical ready rates (approx.): exchange=1.000, news=0.008, event\_calendar=0.000, onchain=0.000, tokenomics=0.000, options=0.000, alternative=1.000, macro=1.000.

## 6. Empirical design

Chronological IS/OOS split at 2026-01-16 (200/200 days). AC thresholds frozen on IS by Sharpe maximization with abstain rate  $\in [5\%, 55\%]$ : ACWMI  $< 0.25$  or  $C < 0.25$  (calibration selects minimal gating). Production WMI threshold 0.2 is never tuned on OOS. Policies: always-long; momentum; mechanism with band content; simple outage; cascade; WMI; ACWMI (IS-frozen). Economic value: annualized return/vol, Sharpe, CRR CE ( $\gamma=2$ ), max DD. **Inference:** circular block bootstrap on OOS daily PnL ( $n_{\text{boot}}=999$ ,  $\text{block}=5$ ) with block-length sensitivity  $\{5,10,21\}$  and a White (2000) reality check; costs 10/25/50 bps on turnover plus perp-funding adjustment. Identification: thin vs thick (content+gating); LOBO with channel decomposition; scarce-world event study; 2017–2026 long-span audit.

### 6.1 Mechanism signals: transparent rule with band content

Directional actions are a deterministic rule on named production calculators, and band *content* enters directly: `macro_tilt` (vintaged VIX/DXY 5d changes: both down = risk-on +1, both up = risk-off -1) and `alt_tilt` (stablecoin 7d net-supply sign, latest obs  $< t$ ). Tilts are forced to 0 when the band is not PIT-ready, so LOBO deletes content, not only gating. Rule:  $R1 \text{ crisis} \wedge \text{cascade\_p} \geq 0.60 \rightarrow \text{short}$  (evidence conjunction);  $R2 \text{ trend} \wedge \text{mom5} > 0 \wedge \text{cascade\_p} < 0.45 \wedge \text{macro\_tilt} \geq 0 \rightarrow \text{long}$ ;  $R2b \text{ range} \wedge \text{both tilts} > 0 \wedge \text{mom5} \geq 0 \rightarrow \text{long}$ ;  $R3 \text{ sign}(\text{mom5})$  with a double-risk-off long veto ( $\rightarrow \text{flat}$ ), ties broken by  $\text{sign}(\text{tilt sum})$ .  $C$  = pairwise sign agreement among  $\{\text{mom5}, \text{flow}, -1_{\{\text{casc} > 0.55\}}, -1_{\{\text{sys} > 55\}}, \text{macro\_tilt}, \text{alt\_tilt}\}$ .

**Calibration audit and engine-input repair (disclosed).** A long-span audit exposed two harness flaws: full-history drawdowns made crisis an absorbing state, and the cascade input map saturated at 0.86 (a constant short that won a bear OOS by construction). Repaired with 60-day windows, volatility-standardized tail intensity, and real RSI/ADX proxies; `cascade_p` is now monotone in realized next-day left tails ( $P(r < -5\%)$  rising 0.039  $\rightarrow$  0.110 across deciles). The repair was driven by the audit, not by OOS performance search.

| component                    | formula                                                                                                                                                                                                                                         | inputs                                                                                                                      | role                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <code>cascade_p</code>       | <code>LiquidationCascadeCalculator.compute_cascade_probability(cluster, capacity, distance)</code>                                                                                                                                              | left-tail cluster intensity from pre-t returns; capacity=6e7; distance by vol regime                                        | crisis / short trigger in R1                                                 |
| <code>systemic</code>        | <code>ContagionRiskCalculator.compute_systemic_risk_score([covar_95, corr, tail_beta])</code>                                                                                                                                                   | asset vs synthetic peer path from pre-t returns                                                                             | sign in consistency C                                                        |
| S (signal integrity)         | <code>clip(hl_factor*(1-0.7*crowding_n)*(0.35+0.65*surprise_n))</code>                                                                                                                                                                          | AlphaDecay half-life on <code>cumsum(returns)</code> , crowding, surprise of last return                                    | ACWMI factor                                                                 |
| C (consistency)              | pairwise sign-agreement among $\{\text{mom5}, \text{flow\_sign}, -1_{\{\text{casc} > 0.55\}}, -1_{\{\text{sys} > 55\}}\}$                                                                                                                       | <code>mom5</code> , VPIN/flow class, <code>cascade_p</code> , <code>systemic</code>                                         | ACWMI factor + AC abstention gate                                            |
| <code>detected_regime</code> | <code>RegimeClassifier.classify_price_regime(RegimeFeatures)</code> with crisis override                                                                                                                                                        | returns, rolling vol, RSI/ADX proxies, <code>cascade_p</code> , <code>vol_regime</code>                                     | R1/R2 branching                                                              |
| <code>mom5</code>            | <code>sign(mean(r_{t-5:t}))</code>                                                                                                                                                                                                              | last 5 pre-t daily returns                                                                                                  | R2/R3 directional fallback                                                   |
| <code>macro_tilt</code>      | +1 if <code>VIX_5d_chg</code> < 0 and <code>DXY_5d_chg</code> < 0; -1 if both > 0; else 0; forced 0 when macro band not ready                                                                                                                   | vintaged <code>macro_timeseries</code> (available_at $\leq t$ ): VIX, DXY                                                   | macro-band content: veto in R2, band-long in R2b, tie-break in R3, sign in C |
| <code>alt_tilt</code>        | <code>sign(stablecoin_net_supply_change_7d)</code> at latest obs $< t$ ; forced 0 when alternative band not ready                                                                                                                               | <code>alternative_timeseries</code> stablecoin 7d net supply (pre-t)                                                        | alternative-band content: band-long in R2b, tie-break in R3, sign in C       |
| signal                       | $R1 \text{ (crisis AND } \text{casc} > 0.60) \rightarrow -1$ ; $R2 \rightarrow +1$ (macro veto); $R2b \rightarrow +1$ (band-driven); $R3 \text{ sign}(\text{mom5})$ with double-risk-off long veto; ties <code>sign(macro_tilt+alt_tilt)</code> | <code>detected_regime</code> , <code>cascade_p</code> , <code>mom5</code> , <code>macro_tilt</code> , <code>alt_tilt</code> | position when not abstaining; R1 requires evidence conjunction               |

Table. Mechanism component definitions (audit).

## 7. Main results

### 7.1 Band content beats momentum out of sample

Always-long loses badly in this bear window (CE  $-1.157$ ); momentum is nearly flat (Sharpe  $0.101$ , CE  $-0.202$ ); the transparent rule with band content earns Sharpe  $0.767$  and CE  $+0.132$ , standing aside on  $7.5\%$  of asset-days via content-driven vetoes. **Headline: mechanism – momentum  $\Delta\text{CE} = 0.334$ ,  $p = 0.034$ ,  $95\%$  CI  $[0.03, 0.68]$  excludes zero.** The two policies share all return-based inputs and differ only in vintaged macro/alternative content, so this contrast isolates compiled band content.

| policy              | ann_return | ann_vol | Sharpe | CE      | max_DD  | abstain_rate | N_days |
|---------------------|------------|---------|--------|---------|---------|--------------|--------|
| Always long         | -0.8164    | 0.5836  | -1.399 | -1.1568 | -0.4661 | 0.0          | 200    |
| Momentum always     | 0.0506     | 0.5009  | 0.101  | -0.2019 | -0.4997 | 0.0          | 200    |
| Thick ungated       | 0.3846     | 0.5012  | 0.767  | 0.132   | -0.4378 | 0.075        | 200    |
| Simple outage rule  | 0.3846     | 0.5012  | 0.767  | 0.132   | -0.4378 | 0.075        | 200    |
| Simple cascade rule | 0.3846     | 0.5012  | 0.767  | 0.132   | -0.4378 | 0.075        | 200    |
| WMI threshold (0.2) | 0.0        | 0.0     | 0.0    | 0.0     | 0.0     | 1.0          | 200    |
| ACWMI (IS-frozen)   | 0.3846     | 0.5012  | 0.767  | 0.132   | -0.4378 | 0.075        | 200    |

Table 1. OOS economic value on real PIT panel (200 days).

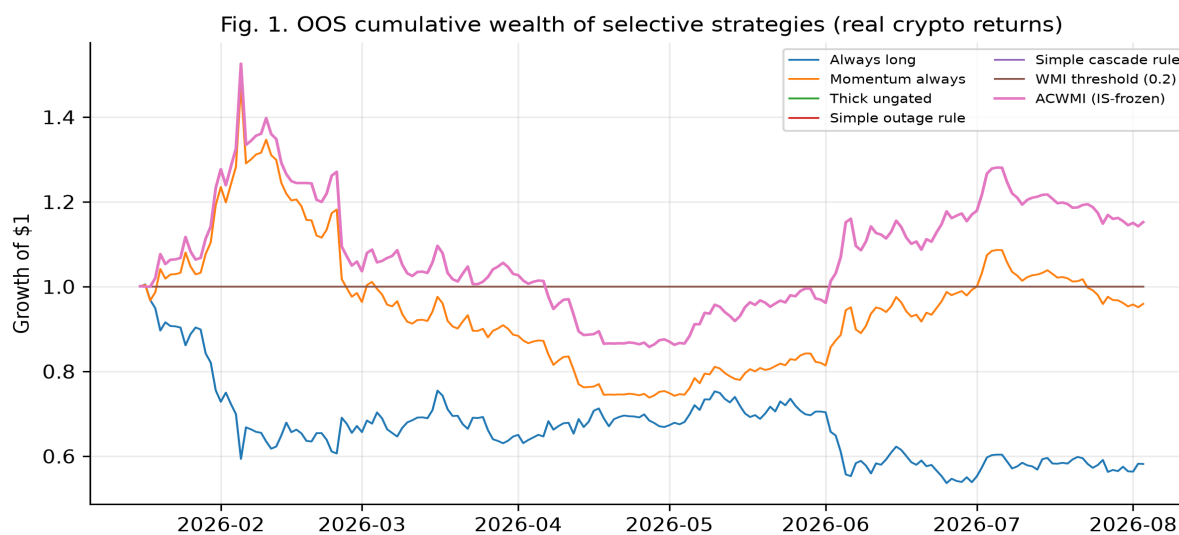


Figure 1. OOS cumulative wealth.



Fig. 2. Economic value of abstention policies (OOS)

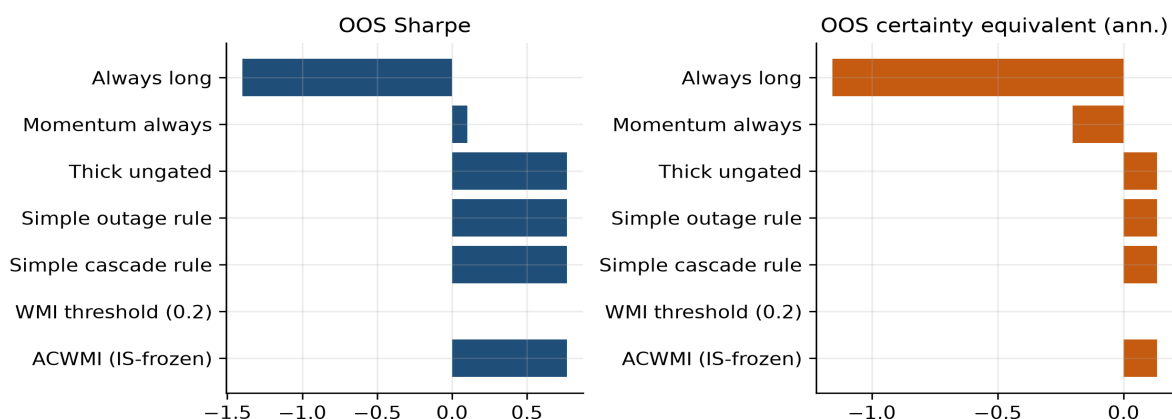


Figure 2. OOS Sharpe and CE by policy.

## 7.2 Thick vs thin worlds

The thin exchange-only observer (content AND gating deleted) abstains 45% and earns CE  $-0.388$ ; the thick world earns  $+0.132$ .  $\Delta\text{Sharpe} = 1.62$  ( $p = 0.044$ , CI excludes zero);  $\Delta\text{CE} = 0.52$  ( $p = 0.22$ , not yet significant at 200 days).

| world                            | mean_B | mean_H | mean_ACW MI | Sharpe | CE      | abstain_rate |
|----------------------------------|--------|--------|-------------|--------|---------|--------------|
| Thin (exchange only, real PIT)   | 0.201  | 0.562  | 0.261       | -0.85  | -0.3883 | 0.454        |
| Thick real PIT (ex+macro+alt...) | 0.356  | 0.688  | 0.368       | 0.767  | 0.132   | 0.075        |
| Thick gated AC (real PIT)        | 0.356  | 0.688  | 0.368       | 0.767  | 0.132   | 0.075        |

Table 2. Thin vs thick on the real PIT archive (OOS).

Fig. 6. Thin vs thick-ungated vs thick-gated worlds (OOS)

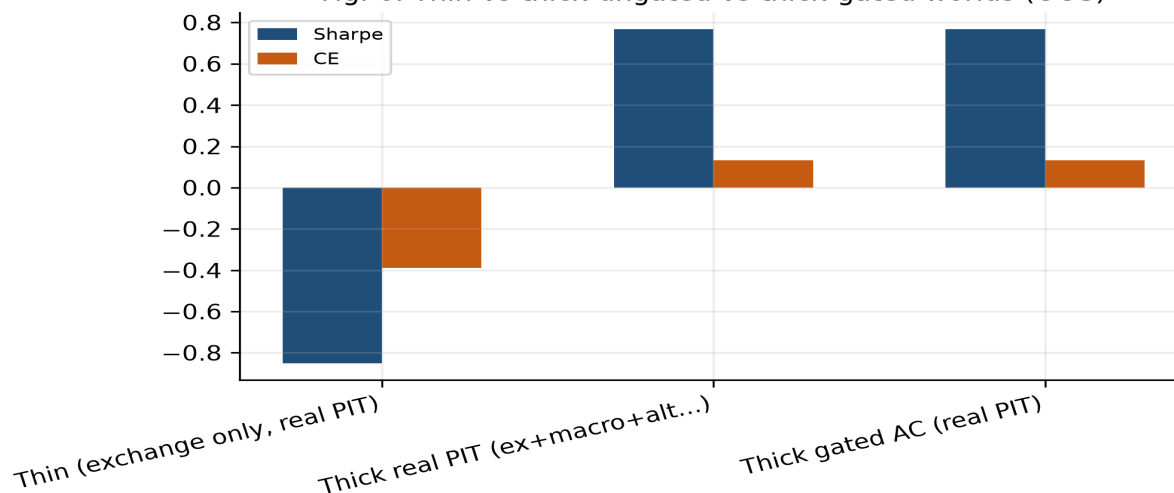


Figure 3. Thin vs thick worlds (real PIT).

## 7.3 Leave-one-band-out: significant, and driven by content

Deleting macro costs  $\Delta\text{CE} -0.418$  ( $p = 0.010$ ); alternative  $-0.402$  ( $p = 0.008$ ); exchange status  $-0.056$  ( $p = 0.394$ ; exchange content is the return data itself). The decomposition shows the loss runs mainly through the *content* channel ( $-0.334$ ,  $p = 0.034$  for both bands) with a smaller significant gating contribution (macro  $-0.066$ ,  $p$

= 0.026; alternative  $-0.040$ ,  $p = 0.044$ ). Band information changes *what* the rule trades, not merely *when* it abstains. Content-only losses coincide across bands because R2b and the double-risk-off veto require both tilts to agree.

| band_dropped      | Sharpe | CE      | abstain_rate | dCE     | p_dCE |
|-------------------|--------|---------|--------------|---------|-------|
| (none)            | 0.767  | 0.132   | 0.075        | 0.0     |       |
| exchange          | 0.65   | 0.076   | 0.158        | -0.056  | 0.394 |
| macro             | -0.08  | -0.286  | 0.06         | -0.418  | 0.01  |
| alternative       | -0.042 | -0.2699 | 0.021        | -0.4019 | 0.008 |
| macro+alternative | -0.025 | -0.2614 | 0.021        | -0.3934 | 0.018 |

Table 3. Leave-one-band-out on durable PIT bands (total effect).

| band              | dCE_total | p_total | dCE_content_only | p_content | dCE_gating_only | p_gating |
|-------------------|-----------|---------|------------------|-----------|-----------------|----------|
| exchange          | -0.056    | 0.394   |                  |           | -0.056          | 0.394    |
| macro             | -0.418    | 0.01    | -0.3339          | 0.034     | -0.0658         | 0.026    |
| alternative       | -0.4019   | 0.008   | -0.3339          | 0.034     | -0.0404         | 0.044    |
| macro+alternative | -0.3934   | 0.018   | -0.3341          | 0.034     |                 |          |

Table 4. LOBO channel decomposition: content vs gating.

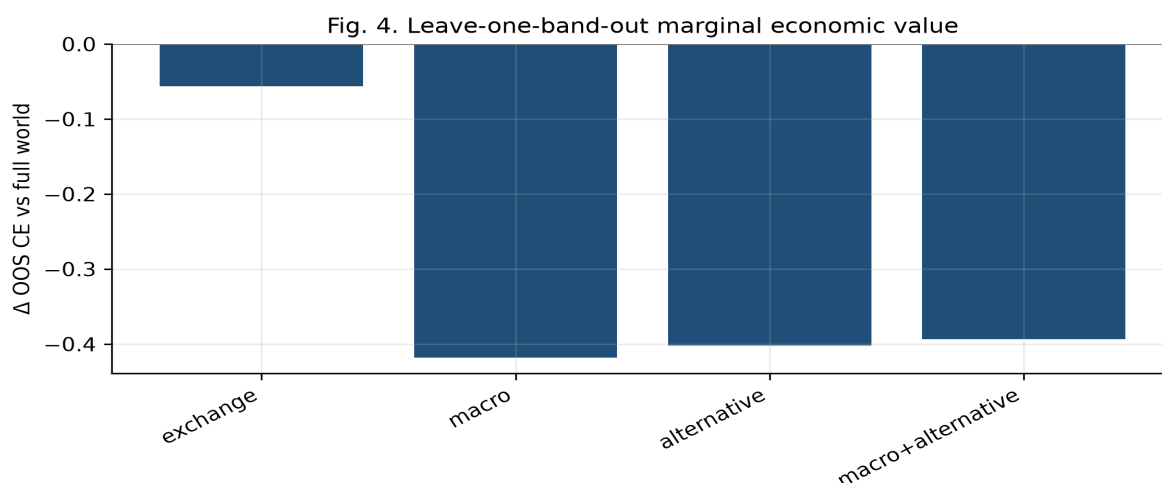


Figure 4. LOBO marginal CE on durable bands.

## 7.4 Long-span external-validity audit (2017–2026)

The identical rule (tilts zero: no band archive before 2025) on BTC/ETH over 3,471 days: annualized return 0.456, Sharpe 0.664, CE  $-0.009$ , positive in 8 of 10 years including 2018 (+2.13 in a  $-1.08$  always-long year) — and no significant advantage over momentum ( $\Delta$ CE 0.017,  $p = 0.55$ ). The return-based core has no hidden alpha; the significant OOS gains load on compiled band content, which exists only where the multi-band archive exists.

| year | n_days | crisis_share | mech_ann_ret | mech_Sharpe | long_ann_ret | long_Sharpe |
|------|--------|--------------|--------------|-------------|--------------|-------------|
| 2017 | 334    | 0.916        | 1.2976       | 1.379       | 3.6928       | 3.916       |
| 2018 | 365    | 0.986        | 2.1252       | 2.639       | -1.0754      | -1.2        |
| 2019 | 365    | 0.752        | 0.8027       | 1.242       | 0.582        | 0.832       |
| 2020 | 366    | 0.54         | 0.6022       | 0.788       | 1.944        | 2.435       |
| 2021 | 365    | 0.968        | 0.4554       | 0.561       | 1.4869       | 1.689       |
| 2022 | 365    | 0.995        | -0.9128      | -1.316      | -0.7852      | -1.075      |
| 2023 | 365    | 0.426        | 0.1539       | 0.374       | 0.8933       | 2.066       |
| 2024 | 366    | 0.71         | 0.1622       | 0.302       | 0.7598       | 1.35        |
| 2025 | 365    | 0.668        | -0.2523      | -0.481      | 0.0926       | 0.165       |

| year | n_days | crisis_share | mech_ann_ret | mech_Sharpe | long_ann_ret | long_Sharpe |
|------|--------|--------------|--------------|-------------|--------------|-------------|
| 2026 | 215    | 0.805        | 0.0247       | 0.049       | -0.5074      | -0.944      |

Table 5. Long-span audit by year (BTC/ETH; crisis\_share is the classifier's label share).

| casc_decile | n    | mean_cascade_p | mean_ret_next | p_tail2 | p_tail5 |
|-------------|------|----------------|---------------|---------|---------|
| 0           | 1989 | 0.2            | 0.00076       | 0.16    | 0.039   |
| 1           | 2357 | 0.309          | 0.00309       | 0.232   | 0.083   |
| 2           | 293  | 0.326          | -0.00126      | 0.256   | 0.072   |
| 3           | 666  | 0.356          | 0.00349       | 0.215   | 0.066   |
| 4           | 660  | 0.403          | 0.00081       | 0.255   | 0.082   |
| 5           | 663  | 0.496          | -0.00056      | 0.267   | 0.11    |

Table 6. cascade\_p calibration: deciles vs realized next-day left tails.

## 7.5 Bootstrap contrasts, costs, and inference robustness

Block-bootstrap ( $999 \times 5$ -day) contrasts: the mechanism – momentum content contrast is significant ( $p = 0.034$ ); contrasts against always-long remain wide at 200 days. Conclusions are stable across block lengths  $\{5, 10, 21\}$ . The White (2000) reality check across the policy menu vs always-long gives  $p = 0.144$  — the menu-wide claim is weaker than the single pre-specified content contrast, disclosed side by side. With 10 bps one-way costs the mechanism keeps Sharpe 0.467 (CE  $-0.019$  vs momentum  $-0.352$ ); at 25 bps both are negative but the ranking is preserved; the perp-funding adjustment moves CE by less than 0.005. B\_hier weight perturbations leave CE unchanged to three decimals.

| contrast                    | dSharpe | dCE    | p_CE  | ci_dCE_05 | ci_dCE_95 |
|-----------------------------|---------|--------|-------|-----------|-----------|
| Thick ungated – Always long | 2.1664  | 1.2887 | 0.246 | -0.9963   | 3.5152    |
| ACWMI – Always long         | 2.1664  | 1.2887 | 0.246 | -0.9963   | 3.5152    |
| ACWMI – Momentum always     | 0.6664  | 0.3339 | 0.034 | 0.028     | 0.6768    |
| Thick ungated – ACWMI       | 0.0     | 0.0    | 1.0   | 0.0       | 0.0       |
| ACWMI – WMI threshold (0.2) | 0.7674  | 0.132  | 0.802 | -1.1986   | 1.3193    |

Table 7. OOS block-bootstrap contrasts.

| policy            | cost_bps | funding              | Sharpe | CE      |
|-------------------|----------|----------------------|--------|---------|
| Thick ungated     | 0.0      | no                   | 0.767  | 0.132   |
| Thick ungated     | 10.0     | no                   | 0.467  | -0.0191 |
| Thick ungated     | 25.0     | no                   | 0.017  | -0.246  |
| Thick ungated     | 50.0     | no                   | -0.725 | -0.6251 |
| Thick ungated     | 10.0     | yes (where archived) | 0.467  | -0.0191 |
| ACWMI (IS-frozen) | 0.0      | no                   | 0.767  | 0.132   |
| ACWMI (IS-frozen) | 10.0     | no                   | 0.467  | -0.0191 |
| ACWMI (IS-frozen) | 25.0     | no                   | 0.017  | -0.246  |
| ACWMI (IS-frozen) | 50.0     | no                   | -0.725 | -0.6251 |
| ACWMI (IS-frozen) | 10.0     | yes (where archived) | 0.467  | -0.0191 |
| Momentum always   | 0.0      | no                   | 0.101  | -0.2019 |
| Momentum always   | 10.0     | no                   | -0.198 | -0.3522 |
| Momentum always   | 25.0     | no                   | -0.646 | -0.578  |
| Momentum always   | 50.0     | no                   | -1.386 | -0.9552 |
| Momentum always   | 10.0     | yes (where archived) | -0.198 | -0.3522 |
| Always long       | 0.0      | no                   | -1.399 | -1.1568 |
| Always long       | 10.0     | no                   | -1.402 | -1.1586 |
| Always long       | 25.0     | no                   | -1.407 | -1.1613 |

| policy      | cost_bps | funding              | Sharpe | CE      |
|-------------|----------|----------------------|--------|---------|
| Always long | 50.0     | no                   | -1.415 | -1.1659 |
| Always long | 10.0     | yes (where archived) | -1.402 | -1.1586 |

Table 8. Transaction-cost and funding sensitivity (OOS).

## 7.6 Selective prediction on a sparse archive

IS calibration selects minimal additional index gating (frozen ACWMI  $< 0.25$  binds on  $\sim 1\%$  of days beyond content stand-asides) and the production WMI  $< 0.2$  rule abstains 100%: quality thresholds are not portable across information-set supports. The measured epistemic calibration penalty is high (ECP = 0.688 at conf  $> 0.7$ , WMI  $< 0.2$ ): the classifier is often confident while the world is thin — the state ECP is designed to flag. All actions bind to named calculator evidence (EAR = 1, UCR = 0).

Implications: (i) compilation quality has first-order economic content and the content channel dominates gating; (ii) mechanism – momentum isolates band content by construction and is significant; (iii) the long-span audit rules out hidden return-rule alpha; (iv) thresholds must be frozen to archive support; (v) remaining contrasts vs always-long are not yet significant at 200 days — point estimates and significance reported separately.

## 8. AI-interface protocol (appendix-grade)

Because the SDF interface concerns agents that condition on  $\mathbf{I}_t^{\text{AI}}$ , the replication package pre-registers a within-model Compiled versus Raw protocol,  $\Delta_m = V_m(\text{Compiled}) - V_m(\text{Raw})$ , with frozen prompts and temperature 0. This is deliberately *not* primary identification. Offline mocks keep the protocol executable without API keys; no LLM trading alpha is claimed.

Offline mock suite on the current OOS window: mean  $\Delta\text{CE}(\text{Compiled-Raw}) = 0.0552$ ; models = mock-compiled-aware, mock-momentum, mock-noisy. Interpret as protocol smoke / calibration check, not as unconditional LLM alpha.

## 9. Robustness, threats, and the JF/RFS agenda

**Right-censored bands.** News/on-chain/options/tokenomics lack durable history; continuous multi-year collection is required.

**Content vintages.** Band-content identification rests on a single 400-day archive window; extend content calendars to multiple regimes.

**Natural outages.** Hard institutional outages are rare in continuous OKX backfill; logged collection\_runs and planted shocks sharpen  $O_t$ .

**Snapshot density.** Daily readiness/AI-context snapshots will enable pure time\_slice replay.

**External validity.** Expand beyond ten liquid names; the 2017–2026 audit anchors the return core only.

**Multiple testing.** The headline contrast is a single pre-specified comparison; the menu-wide reality check ( $p = 0.144$ ) is reported alongside; add stationary-bootstrap confirmation.

**Timing / ACWMI disclosure.** Features for earning  $r_t$  are evaluated at  $(t-1)$  23:59. Live API bundles disclose  $\text{acwmi\_input\_source} \in \{\text{paper\_engines}, \text{production\_proxy}\}$ ; paper empirics use engine (S,C).

## 10. Conclusion

Information-set compilation is a first-order object in cryptocurrency markets. This paper develops the RCA-WM / ACWMI theory as a proposition chain — compilation  $\neq$  feature expansion, an SDF interface, reconstruction bounds, world-conditional abstention, ACWMI monotonicity, and LOBO value with a content/gating decomposition — and instantiates every object on a real multi-band PIT archive. Vintage-safe macro and stablecoin-flow content carries significant OOS economic value ( $\Delta\text{CE}$  vs momentum 0.334,  $p = 0.034$ ; LOBO macro/alternative  $p = 0.010/0.008$  with the content channel dominant), while the long-span audit shows the return-based rule itself has no hidden alpha — exactly the pattern compilation theory predicts. The path to a final submission is institutional: deepen vintaged histories, log outages, snapshot daily, and expand the cross-section — without abandoning the formal apparatus.

## Appendix A. Notation

| Symbol                | Meaning                                     |
|-----------------------|---------------------------------------------|
| $O_{j,t}$             | Epistemic observation                       |
| $B_t, U_t, H_t$       | Breadth, stability, honesty                 |
| $WMI_t / ACWMI_t$     | Production / regime-conditional index       |
| $\Pi_t$               | Compilation operator                        |
| ECP / MIG             | Calibration penalty / band information gain |
| $\Phi_t$ ; EAR/UCR/EV | Explanation set and metrics                 |
| $O_t$                 | Availability shock (DAG)                    |

Table A1. Core notation.

## Appendix B. Evidence bands and laboratory map

The laboratory organizes collectors into evidence bands that feed readiness and AI context. Table B1 lists band-level inventory used by the paper's readiness weights; Table B2 samples logic modules.

| Band            | Module               | Required | Asset weight | Role                                         |
|-----------------|----------------------|----------|--------------|----------------------------------------------|
| exchange        | exchange_data        | Yes      | 0.3          | Microstructure, prices, book, funding, basis |
| macro           | macro_data           | Yes      | 0.05         | Macro vintages / rates / liquidity proxies   |
| news            | news_data            | Yes      | 0.15         | News and narrative evidence                  |
| event_calendar  | event_calendar_data  | Yes      | 0.1          | ETF / unlock / event calendar                |
| onchain         | onchain_data         | Yes      | 0.15         | On-chain activity / TVL proxies              |
| tokenomics      | tokenomics_data      | Yes      | 0.1          | Unlocks / supply-side tokenomics             |
| options         | options_data         | Yes      | 0.1          | Options surface / gamma walls                |
| alternative     | alternative_data     | No       | 0.05         | Alt data (stablecoins, GitHub, trends)       |
| perpetual_dex   | perpetual_dex_data   | No       |              | Perp DEX funding / OI                        |
| onchain_address | onchain_address_data | No       |              | Address-level on-chain labels                |
| dex_liquidity   | dex_liquidity_data   | No       |              | DEX TVL / tick liquidity                     |
| gas_network     | gas_network_data     | No       |              | Gas / network congestion                     |
| governance      | governance_data      | No       |              | DAO governance events                        |

Table B1. Evidence bands (excerpt).

| Module (1)                 | Module (2)          | Module (3)               |
|----------------------------|---------------------|--------------------------|
| ai_market_context          | alpha_decay         | anomaly_detection        |
| asset_readiness            | contagion_risk      | cross_asset_analysis     |
| cross_venue_arbitrage      | defi_stress         | depth_regime             |
| event_probability          | exchange_comparison | feature_standardization  |
| flow_decomposition         | funding_rate_model  | holder_behavior_analysis |
| liquidation_cascade        | liquidity_analysis  | liquidity_regime         |
| logic_pipeline             | macro_context       | market_breadth           |
| market_sentiment_composite | market_structure    | miner_pressure           |
| narrative_regime           | news_sentiment      | onchain_lead_lag         |
| pipeline_latency           | portfolio_risk      |                          |

Table B2. Logic modules (excerpt, 3-column grid).

| Theory object            | EvoQuant implementation                       | Role in paper           |
|--------------------------|-----------------------------------------------|-------------------------|
| Market state compilation | data_layer collectors + logic_pipeline DAG    | Core system             |
| Breadth                  | 43 domains / 13 audit bands / asset_readiness | Hierarchical B_hier     |
| Stability                | pipeline_latency freshness summary            | U in WMI/ACWMI          |
| Honesty                  | quality_flag + exclusion/contamination        | Continuous H_cont       |
| Signal integrity         | alpha_decay half-life/crowding/surprise       | S factor                |
| Consistency              | cross-band directional agreement              | C factor                |
| Mechanism engines        | regime/cascade/contagion/flow/vol             | Psi_mech                |
| Degraded operation       | DegradationManager levels                     | Resilient honesty       |
| Abstention               | should_ai_abstain + AC policy                 | State-dependent refusal |

Table B3. Theory-to-implementation map.

## Appendix C. Additional empirics

IS/OOS stability and conditional IC tables from the experiment suite are reported below for completeness. They complement the main CE/Sharpe horse-race and should be read under the same frozen-threshold protocol.

| split | policy | Sharpe  | CE      |
|-------|--------|---------|---------|
| IS    | WMI    | 0.0     | 0.0     |
| IS    | ACWMI  | -0.2603 | -0.4467 |
| IS    | Mom    | -0.2153 | -0.4292 |
| OOS   | WMI    | 0.0     | 0.0     |
| OOS   | ACWMI  | 0.7674  | 0.1320  |
| OOS   | Mom    | 0.1009  | -0.2019 |

Table C1. IS/OOS stability.

| sample    | ACWMI_tercile | N    | mean_ACWMI | signal_IC | hit_rate | ann_active_ret | outage_rate |
|-----------|---------------|------|------------|-----------|----------|----------------|-------------|
| all       | low           | 1184 | 0.518      | 0.00515   | 0.575    | 1.8812         | 0.247       |
| all       | mid           | 1183 | 0.597      | 0.00281   | 0.568    | 1.025          | 0.006       |
| all       | high          | 1183 | 0.706      | -0.00142  | 0.495    | -0.5182        | 0.0         |
| no_outage | low           | 1084 | 0.545      | 0.00367   | 0.534    | 1.3408         | 0.0         |
| no_outage | mid           | 1082 | 0.606      | 0.00227   | 0.571    | 0.8288         | 0.0         |
| no_outage | high          | 1084 | 0.712      | -0.00106  | 0.504    | -0.3878        | 0.0         |

Table C2. Conditional IC.

| date       | ready | stale | missing | freshness   | klines  | readiness | ai_ctx  |
|------------|-------|-------|---------|-------------|---------|-----------|---------|
| 2025-07-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2025-08-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2025-09-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2025-10-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2025-11-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2025-12-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2026-01-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2026-02-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2026-03-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2026-04-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2026-05-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |
| 2026-06-01 | 0     | 0     | 11      | unavailable | missing | missing   | missing |

Table C3. time\_slice monthly grid (summary; analytics snapshots remain sparse historically).

## Appendix D. Reproducibility

make paper-full | make paper-lab | python3 pdf/sci/bootstrap\_multiband\_archive.py | build\_pit\_archive.py | run\_pit\_jf\_experiments.py | generate\_full\_manuscript\_pdf.py. Artifacts: pdf/data/pit\_multiband\_panel.csv; pdf/sci/main\_jf\_rfs.tex; pdf/cn/main\_cn\_jf.md.

*Draft status: structurally complete working paper (theory + real PIT empirics + agenda). Acceptance at JF/RFS still requires deeper vintaged histories, logged outages, broader cross-section, and journal-format polishing—without dropping the formal apparatus again.*



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